Lab 7: Fuzzing for Input Validation Bugs

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Files: `income_tax_parser.py` (fixed module) and `test_income_tax_parser.py` (Hypothesis tests). Screenshots SS1 to SS4 follow on the next pages.

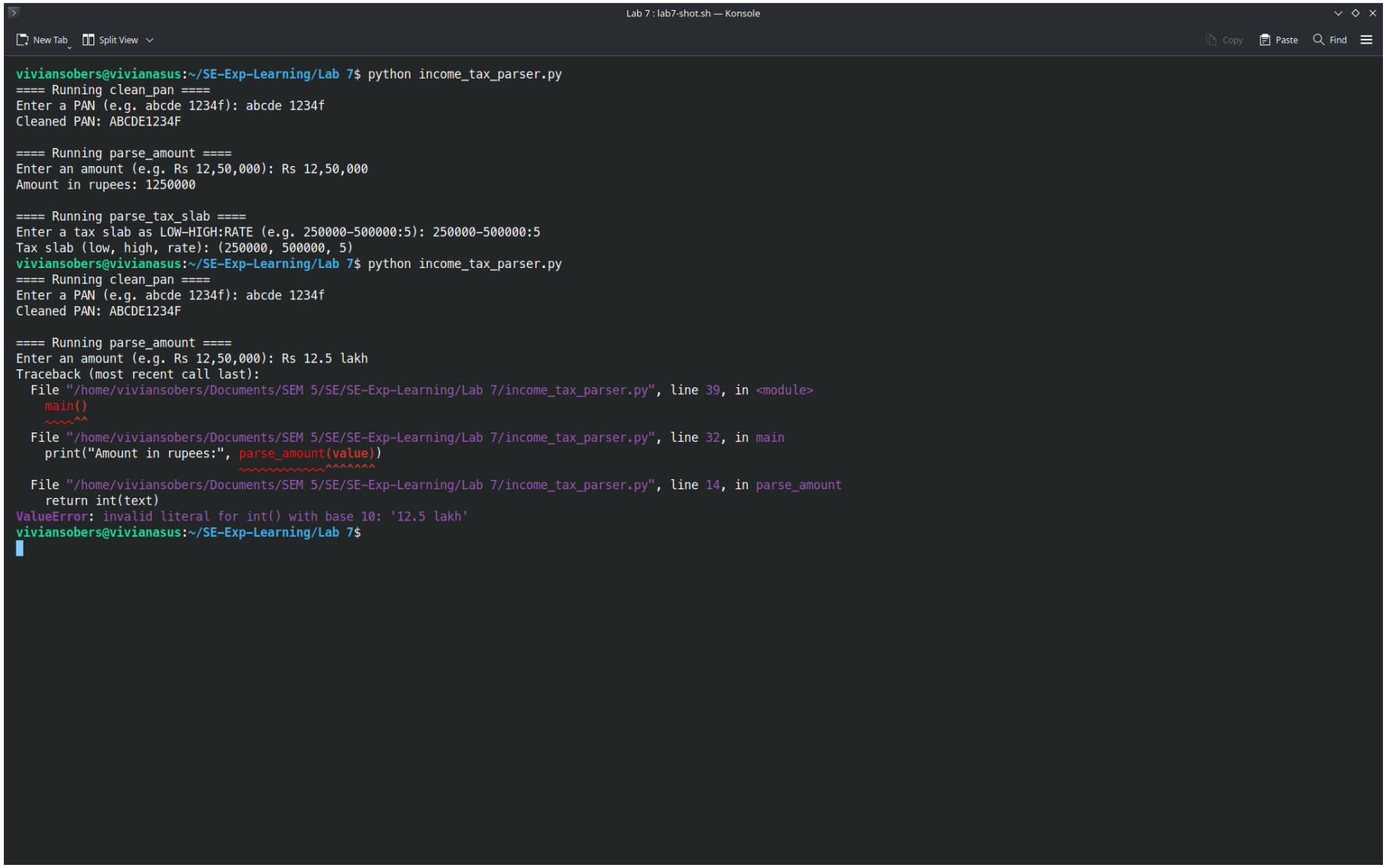
Bugs found and how they were fixed

Function	What crashed or went wrong	Fix
`clean_pan`	None, numbers or bytes raised `AttributeError`. Any string was accepted, so `""` or `"hello"` came back as a PAN, and `"abcdı1234f"` became ABCDI1234F because `"ı".upper()` is `"I"`.	Non-strings return None. All whitespace is removed, non-ASCII input is rejected, and the result must be 5 letters, 4 digits and 1 letter.
`parse_amount`	Non-strings raised `AttributeError`. Text such as `"12.5 lakh"` or `""` raised `ValueError`. `int()` also accepted signs, underscores and non-ASCII digits, and a string of more than 4,300 digits raises ValueError.	Non-strings return None. An optional leading `Rs`, `Rs.` or ₹ is removed along with commas and spaces, and only 1 to 15 ASCII digits are converted.
`parse_tax_slab`	Non-strings raised `AttributeError`. A missing or extra `:` or `-` raised `ValueError` on unpacking, and non-numbers raised `ValueError` in `int()`. A slab with low above high, or a rate over 100, was accepted.	The whole line must match `LOW-HIGH:RATE` with ASCII digits, low must not exceed high and the rate must be 0 to 100. Anything else returns None.
`main`	Any of the crashes above stopped the program.	Prints `Invalid PAN`, `Invalid amount` or `Invalid tax slab` and carries on.

The tests: three Hypothesis tests feed each function strings, None, integers, floats, bytes and lists (500 examples each) and fail on any exception or a wrong return type. Three more generate valid PANs, amounts and slabs and check the exact result, so a fix that rejected everything would still fail. The seventh test covers the non-ASCII PAN case.

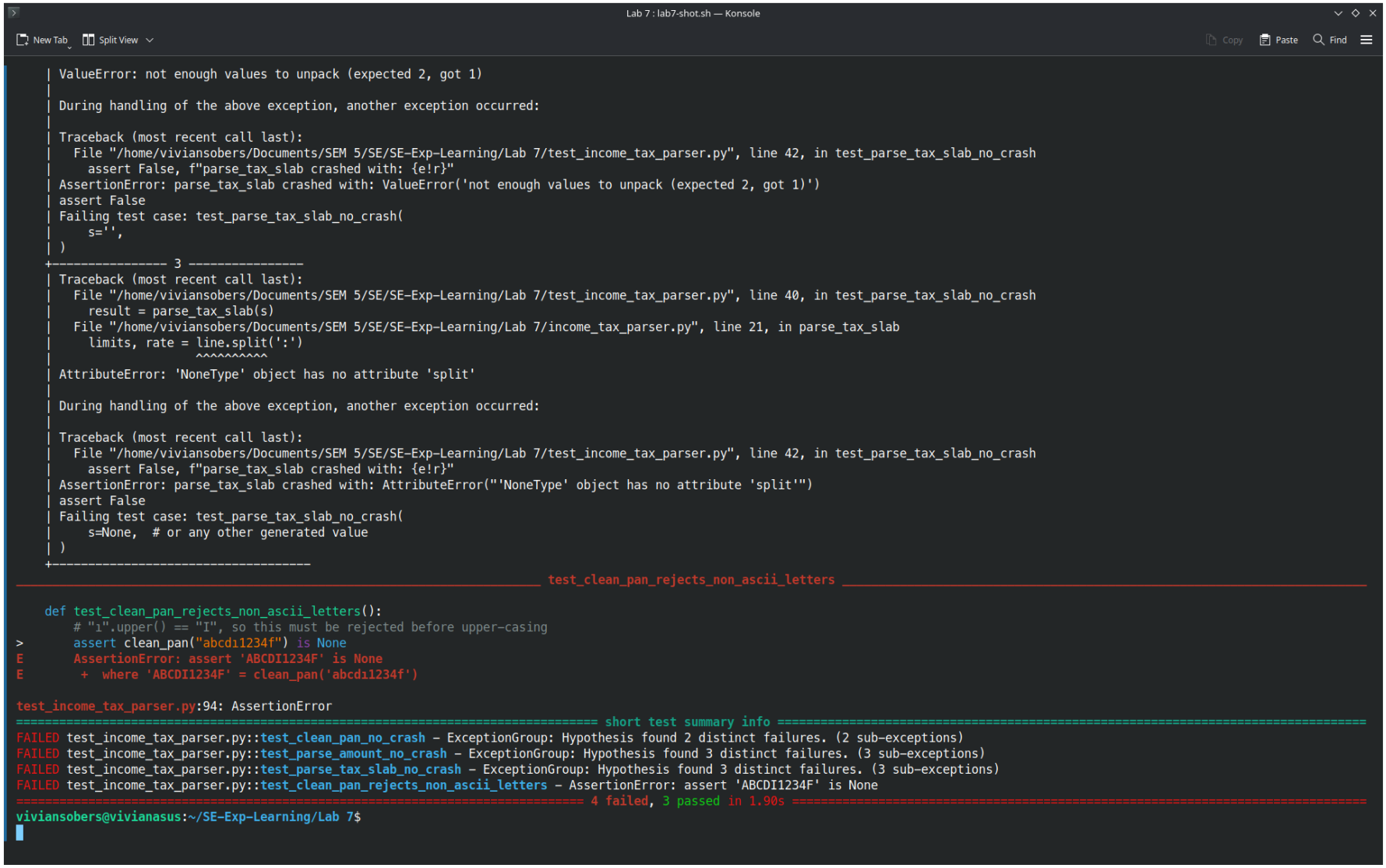
SS1: `python income_tax_parser.py on the buggy module`

The first run uses well-formed input and works. The second run types the amount `Rs 12.5 lakh`, and `parse_amount` crashes with a ValueError, so the program never reaches the tax slab.



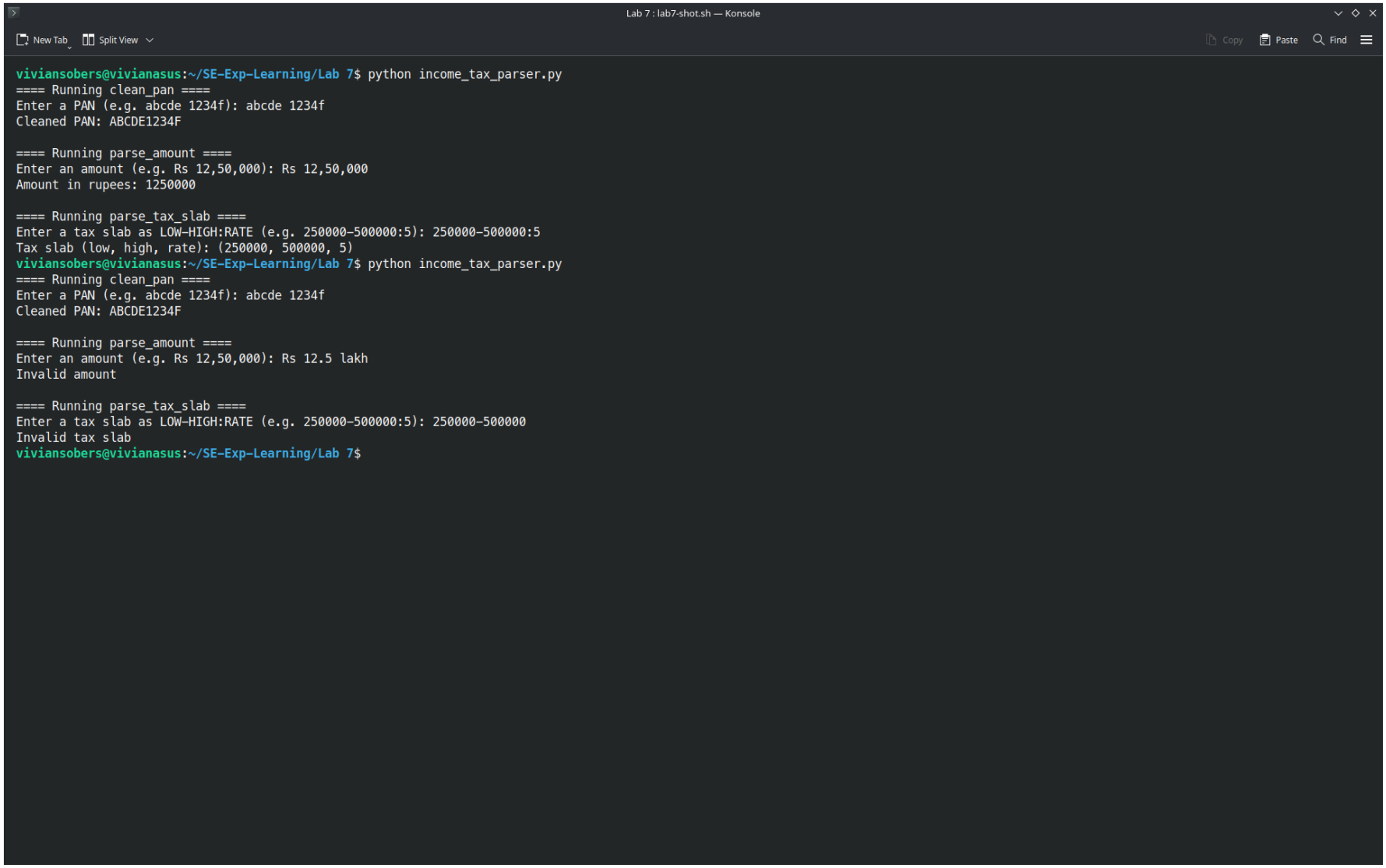
SS2: `python -m pytest test_income_tax_parser.py on the buggy module`

4 of 7 tests fail. Hypothesis found 2 distinct crashes in clean_pan, 3 in parse_amount and 3 in parse_tax_slab, and clean_pan also accepted a PAN containing a non-ASCII letter.



SS3: `python income_tax_parser.py on the fixed module`

The same two runs as SS1. Valid input gives the same results as before. The malformed amount and the slab with no rate are reported as invalid, and the program finishes instead of crashing.



SS4: `python -m pytest test_income_tax_parser.py on the fixed module`

All 7 tests pass.

